

Function Point Analysis

Main technologies used:

- Backend: Node.js/JavaScript (60%)
- Frontend: React/JavaScript (30%)
- Database: PostgreSQL (10%)

1. Convert Function Points to SLOC

Function Points: 330 AFP

Language Conversion Factors:

- JavaScript/Node.js: 47 LOC/FP
- React/JSX: 46 LOC/FP
- SQL: 13 LOC/FP

Weighted Average = $(0.60 \times 47) + (0.30 \times 46) + (0.10 \times 13)$
= $28.2 + 13.8 + 1.3 = 43.3$ LOC/FP

Total Estimated SLOC = $330 \times 43.3 = 14,289 \approx 14.3$ KSLOC

2. Determine Scale Drivers

Scale Factor | Rating | Value | Justification

PREC (Precedentedness) | Nominal | 3.72 | Healthcare platforms exist but microservices approach is moderately novel

FLEX (Development Flexibility) | High | 2.03 | Academic project with flexible requirements

RESL (Architecture/Risk Resolution) | Nominal | 3.29 | Clear architecture but student-level risk analysis

TEAM (Team Cohesion) | High | 2.83 | Small, cohesive student team

PMAT (Process Maturity) | Low | 6.24 | Limited organizational process maturity

Scale Factor (SF) = 18.11

Scale Exponent (E) = $B + 0.01 \times SF = 0.91 + 0.01 \times 18.11 = 1.09$

3. Effort Multipliers (EM)

Effort Adjustment Factor (EAF) = 1.85

4. COCOMO II Estimation – Early Design Model

Model Parameters:

- COCOMO II Model: Early Design
- A (constant): 2.94
- EAF (Effort Adjustment Factor): 1.85
- Scale Exponent (E): 1.09
- Size: 14.3 KSLOC

Calculations:

Effort = $2.94 \times (14.3)^{1.09} \times 1.85 = 92.3$ person-months

Schedule (TDEV) = $3.67 \times (92.3)^{0.316} = 14.2$ months

Team Size = $92.3 / 14.2 = 6.5$ developers

5. Academic Project Context Adjustment

Team Size → 3 developers

Time Duration → 5 months

Total Effort → 16 person-months

Actual Productivity → $14,289 / 16 = 893$ SLOC/person-month

6. Reverse Engineering the COCOMO II Model

Feasible SLOC = $16 \times 600 = 9,600$ SLOC

Equivalent Function Points = $9,600 / 43.3 \approx 222$ FP

7. Final Estimates

Metric | COCOMO II Estimate | Academic Reality

Effort | 92.3 person-months | 16 person-months

Duration | 14.2 months | 5 months

Team Size | 6.5 developers | 3 developers

Productivity | 155 SLOC/person-month | 893 SLOC/person-month